Dublin Business School

Assessment Template

Computing

Assessment:

Item	Description
Module Title:	Reinforcement Learning
Module Code:	B9AI105
Module Lecturer (s):	Dr. Anesu Nyabadza
Programme/Cohort:	MSc Artificial Intelligence
Method of Assessment:	Artifact demonstration
Percentage (%) Weighting:	60%
MIMLOs being assessed:	1, 2, 3, 4
Assessment Number:	1
Individual/Group:	Group
Issue Date:	Week 5
Submission Date:	Week 11
Feedback Date:	Week 12
Feedback Strategy:	In class and Moodle

Assignment Task

Task 1 : Artifact presentation

Solve a real-world problem using Reinforcement Learning models. Building from CA 1, students need to deploy their RL solution developed in CA 1 and demonstrate its real-world application (e.g., in an app, controlling a physical robot/electronics, website or software).

This task requires each team (3-5 student) to present a demo of their working solutions and answer questions about how the solution works, limitations, ethics and defend their decision making.

The goal is to demonstrate real-world application of the solution above the report, theory and code. The goal is to evaluate the solution qualitatively (in real application) for example examining a real robotic car move across the room with obstacles (or a simulated version of the car).

- Teams must clearly explain the motivation for the chosen problem, why RL is an appropriate method, and what impact the solution could have in real-world settings.
- The teams must mention challenges/differences between theoretical solution vs deployed solution.
- The work must show comprehension of RL concepts such as agents, environments, states, actions, rewards, value estimation, exploration strategies, and policy optimisation.
- The methodology should be described in detail, including:

(100 marks)

Overall marks will be awarded according to the following criteria:

Presentation skills	25%
Working deployment of RL solution	25%
Ethics	20%
Limitations	15%
Teamwork	15%

Resources

Software or tools required/ useful: Python

Templates, files, data tables provided by the lecturer: Moodle resources

Highlight the Appropriate Selection as per the programme Module and Assessment Document

1	2	3	4	5
NO AI	**AI-ASSISTED IDEA GENERATION AND STRUCTURING**	**AI-ASSISTED EDITING**	**AI TASK COMPLETION, HUMAN EVALUATION**	**FULL AI**
The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their knowledge, understanding, and skills. **AI must not be used at any point during the assessment.**	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. **No AI content is allowed in the final submission.**	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can be created using AI. **AI can be used, but your original work with no AI content must be provided in an appendix.**	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output. **You will use AI to complete specified tasks in your assessment. Any AI created content must be cited.**	AI should be used as a 'co-pilot' in order to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. **You may use AI throughout your assessment to support your own work and do not have to specify which content is AI generated**

Grading Criteria		Evaluative Criteria/Performance Level Descriptors (what does performance at this level look like?)			
Scale	**Grade Band**	Criterion 1 - Presentation skills and Teamwork	Criterion 2 – Working deployment of RL solution	Criterion 3 – Ethics	Criterion 4 – Limitations
Exceptional	**90-100%**	Clear, confident, professional delivery. Logical structure, excellent visuals. All members contribute smoothly and demonstrate deep understanding. Handles questions expertly.	Fully functional, real-time deployment. Demonstrates agent acting in environment with measurable rewards. Robust, reliable behaviour. Clear link to CA1 solution.	Identifies all relevant ethical concerns (safety, bias, transparency, data use). Discusses mitigation strategies and societal impact convincingly.	Clear, honest evaluation of limitations: technical, real-world constraints, data, hardware, agent behaviour. Suggests realistic improvements and future work.
Outstanding	**80-89%**	Strong delivery with good structure and visuals. Most members contribute effectively. Answers questions well with minor gaps.	Functional deployment with minor issues. Demonstrates learning and real-world interaction. Mostly reliable behaviour.	Strong awareness of ethical implications. Mitigations discussed, some aspects less developed.	Good identification of limitations. Improvement ideas present, minor gaps.
Excellent	**70-79%**	Good delivery and clear explanation. Some members more dominant than others. Answers questions adequately.	Working prototype with visible agent–environment interaction. Some instability or limited performance.	Identifies main ethical issues and provides reasonable discussion.	Main limitations identified and explained.

Very Good	**60-69%**	Competent delivery but pacing/structure may be uneven. Team participation mostly balanced. Responses to questions satisfactory.	Partial deployment. Core functionality works but with inconsistencies, bugs, or reduced realism.	Acknowledges some ethical concerns but limited depth or mitigation.	Some limitations mentioned, limited analysis.
Good	**50-59%**	Basic delivery with limited structure or weaker visuals. Uneven contribution across team. Answers questions inconsistently.	Basic demo or simulation only. Performance limited, learning not clearly evident.	Minor mention of ethics with minimal explanation.	Basic mention of limitations, little justification.
Sufficient	**40-49%**	Presentation difficult to follow. Limited teamwork. Answers to questions vague or incomplete.	Minimal functionality. Little evidence of RL behaviour beyond code execution.	Very basic ethical statement; lacks relevance to solution.	Minimal or unclear description of limitations.
Somewhat/ Unfulfilled	**30-39%**	Poor delivery, unclear explanation. Minimal teamwork. Unable to answer most questions.	Non-working or mostly theoretical demonstration.	Little or no ethical consideration.	Major limitations ignored.
Unfulfilled	**0-29%**	No coherent presentation. No teamwork demonstrated. Unable to answer questions.	No deployment shown.	No ethics discussed.	No acknowledgement of limitations.
MIMLO Alignment (see Module Guide for Minimum Intended Learning Outcomes for Module)					
MIMLO to which criterion is aligned:		1, 2	1, 2, 4	1, 2, 3, 4	3, 4

General Requirements for Students:

1. All assignments must be submitted no later than the stated deadline (date and time).
2. Assignments submitted after the latest deadline specified (including any approved extension deadline) are considered late and penalised according to the Quality Assurance Handbook (QAH) **Part B Section 5.2.2.6** as follows:
    a. A penalty of 2 marks will be applied per day or part thereof (including weekends and public holidays) for an ongoing failure to submit beyond the submission deadline.
    b. An examiner has the right to refuse to mark the assignment if the submission instructions have not been observed.
    c. Where a late assessment is submitted within 14 days of the deadline, and is of a passing standard, the late penalty is capped (such that the minimum grade that can be awarded is 40% for the late submission).
    d. Where a late assessment is submitted more than 14 days after the deadline, it will receive 0%. The lecturer may, at their discretion, review the submission for feedback.
    e. Where the assessment is undertaken in a group, the piece of work should be submitted in its complete entirety, and any penalty for late submission incurred applies to all group members.
3. Extensions to assignment submission deadlines will not be granted, other than in exceptional circumstances. To apply for an extension please go to https://students.dbs.ie/dashboard/SCCM and open a ticket.
4. All relevant provisions of the Assessment Regulations must be complied with (see QAH B.5).
    a. Students are required to refer to the assessment regulations in their Programme Handbook, and on the Student Website.
    b. Dublin Business School penalises students who engage in academic impropriety (i.e. plagiarism, collusion and/or copying, ghost writing/ essay mills, improper use of Generative Artificial Intelligence software).
        i. Refer to the College's Generative AI Guidelines HERE for further information.
    c. Guides on referencing are available on the Library website: https://libguides.dbs.ie/referencing
    d. Text-matching analysis software is integrated in Moodle to generate a report regarding the degree of text-matching in a submission.
5. Students are required to retain a copy of each assignment submitted, until the issuing of a transcript indicating the mark awarded and the closure of the Appeal period (2 weeks following the release of final results).
    a. Results can only be appealed following the release of final results, and the Appeal form must be submitted to the Exams Office within the Appeal period.
    b. An appeal must be based on valid grounds (see the Appeals Policy QAH B.3.5), dissatisfaction with a grade is not sufficient grounds for an appeal.
    c. Assignments must be appropriately packaged and presented.
    d. All assignments should be submitted to your subject/course page on Moodle by the deadline date.

e. Where a submission involves digital media (i.e formats other than Word, Powerpoint or PDF), it is the submitting students' responsibility to ensure the media is appropriately labelled, fully working and they must retain a copy.

f. Components of an assessment which are not included in the final submission cannot normally be subsequently accepted for grading. It is the student's responsibility to ensure their file is uploaded correctly.

g. Include an electronic **cover sheet** with the following details to the front of the assignment (see below)

6. Assignments that *breach* the word count requirements will be penalised. *There is a 10% discretion, either way, applicable in terms of word count.*

7. When you submit your assignment you will be asked to click on a button which will declare the following:

By ticking this box I am confirming that this assignment/exam is all my own work. Any sources used have been referenced. *I have read the College rules regarding plagiarism in the QAH Part B Section 3 and understand that penalties will be applied accordingly if work is found not to be my own. All work uploaded is submitted via Ouriginal, whereby a text-matching report will show any similarities with other texts.*